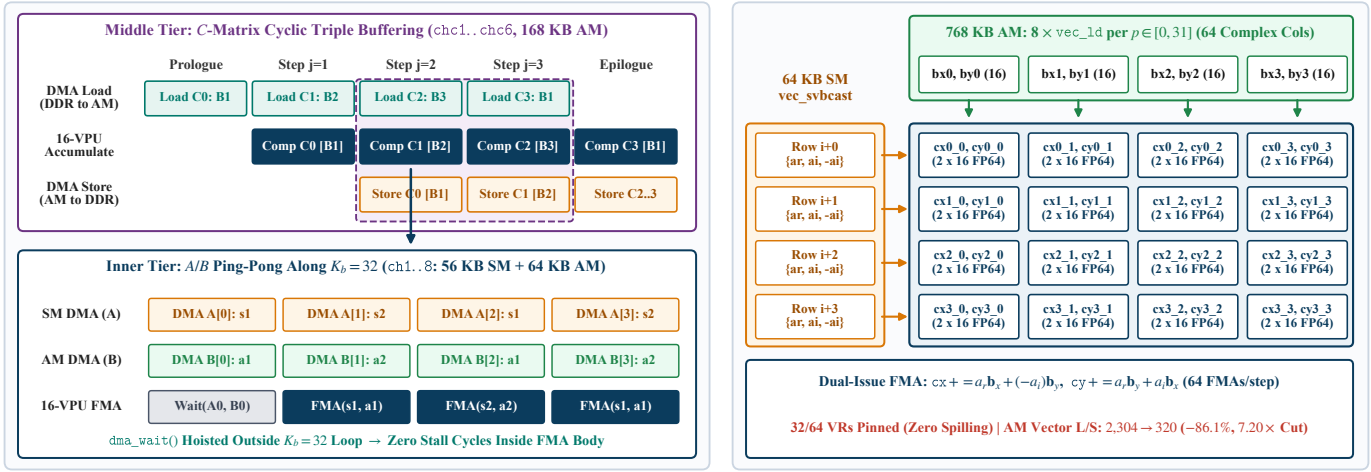

(a) 14-Channel Asynchronous DMA Space-Time Pipeline
Middle Tier: C-Matrix Cyclic Triple Buffering (chc1..chc6, 168 KB AM)
Prologue
Step j=1
Step j=2
Step j=3
Epilogue
DMA Load (DDR to AM)
Load C0: B1
Load C1: B2
Load C2: B3
Load C3: B1
16-VPU Accumulate
Comp C0 [B1]
Comp C1 [B2]
Comp C2 [B3]
Comp C3 [B1]
DMA Store (AM to DDR)
Store C0 [B1]
Store C1 [B2]
Store C2..3
Inner Tier: A/B Ping-Pong Along $K_b = 32$ (ch1..8: 56 KB SM + 64 KB AM)
SM DMA (A)
DMA A[0]: s1
DMA A[1]: s2
DMA A[2]: s1
DMA A[3]: s2
AM DMA (B)
DMA B[0]: a1
DMA B[1]: a2
DMA B[2]: a1
DMA B[3]: a2
16-VPU FMA
Wait(A0, B0)
FMA(s1, a1)
FMA(s2, a2)
FMA(s1, a1)
dma_wait() Hoisted Outside $K_b = 32$ Loop → Zero Stall Cycles Inside FMA Body
(b) 4x4 Register-Resident Dual-FMA Micro-Kernel (32 VRs)
768 KB AM: 8 × vec_ld per $p \in [0, 31]$ (64 Complex Cols)
64 KB SM vec_svbcast
bx0, by0 (16)
bx1, by1 (16)
bx2, by2 (16)
bx3, by3 (16)
Row i+0 {ar, ai, -ai}
Row i+1 {ar, ai, -ai}
Row i+2 {ar, ai, -ai}
Row i+3 {ar, ai, -ai}
cx0_0, cy0_0 (2 x 16 FP64)
cx0_1, cy0_1 (2 x 16 FP64)
cx0_2, cy0_2 (2 x 16 FP64)
cx0_3, cy0_3 (2 x 16 FP64)
cx1_0, cy1_0 (2 x 16 FP64)
cx1_1, cy1_1 (2 x 16 FP64)
cx1_2, cy1_2 (2 x 16 FP64)
cx1_3, cy1_3 (2 x 16 FP64)
cx2_0, cy2_0 (2 x 16 FP64)
cx2_1, cy2_1 (2 x 16 FP64)
cx2_2, cy2_2 (2 x 16 FP64)
cx2_3, cy2_3 (2 x 16 FP64)
cx3_0, cy3_0 (2 x 16 FP64)
cx3_1, cy3_1 (2 x 16 FP64)
cx3_2, cy3_2 (2 x 16 FP64)
cx3_3, cy3_3 (2 x 16 FP64)
Dual-Issue FMA: cx += $a_r \mathbf{b}_x + (-a_i)\mathbf{b}_y$, cy += $a_r \mathbf{b}_y + a_i \mathbf{b}_x$ (64 FMAs/step)
32/64 VRs Pinned (Zero Spilling) | AM Vector L/S: 2,304 → 320 (−86.1%, 7.20× Cut)